

Iris Dataset - Exploratory Data Analysis (EDA)

Project Overview

This project performs Exploratory Data Analysis (EDA) on the Iris Dataset using Python libraries such as Pandas, Matplotlib, and Seaborn.

The goal of this project is to understand the dataset, identify patterns, analyze feature relationships, and visualize the data using various statistical and graphical techniques.

Dataset Information

The Iris Dataset contains 150 flower samples belonging to three different species:

- Iris-setosa
- Iris-versicolor
- Iris-virginica

Features

- SepalLengthCm
 - SepalWidthCm
 - PetalLengthCm
 - PetalWidthCm
 - Species
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Tools and Libraries Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
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Tasks Performed

1. Data Loading

- Imported the Iris dataset into Pandas DataFrame.

2. Data Inspection

- Checked dataset shape
- Viewed data types
- Displayed summary statistics

3. Data Cleaning

- Checked for missing values
- Checked for duplicate records

4. Exploratory Data Analysis

- Species distribution analysis
- Histograms
- Boxplots
- Pairplots
- Correlation Heatmap
- Scatter Plot Analysis

5. Statistical Analysis

- Mean
 - Standard Deviation
 - Minimum and Maximum Values
 - Feature Correlations
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Key Findings

- The dataset contains 150 records and 5 features.
 - No missing values were found.
 - No duplicate records were found.
 - All three species are equally distributed.
 - Petal Length and Petal Width show a strong positive correlation.
 - Petal-related features are highly useful for distinguishing flower species.
 - Iris-setosa is clearly separable from the other species.
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Visualizations Generated

- Histograms
 - Boxplots
 - Pairplots
 - Correlation Heatmap
 - Scatter Plot
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Conclusion

The Iris dataset is clean, balanced, and suitable for machine learning applications. Exploratory Data Analysis revealed strong relationships between petal measurements and flower species, making these features important for classification tasks.

Project Structure

```
|— Iris.csv
|— EDA_Iris.ipynb
|— README.md
|— screenshots/
|   |— histogram.png
|   |— boxplot.png
|   |— pairplot.png
|   |— heatmap.png
```

Author

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